

		d^{-}
21	D	Resistance of voltmeter is equal to resistance of $P = R_p$ Voltmeter and P are in parallel, so effective resistance $= 0.5R_p$ By potential divider principle, $V = \frac{0.5R_p}{0.5R_p + R_Q} E$ $\frac{6.0}{9.0} = \frac{0.5R_p}{0.5R_p + R_Q}$ $\frac{2}{3} = \frac{0.5R_p}{0.5R_p + R_Q}$ $0.5R_p + R_Q = 0.75R_p$ $0.25R_p = R_Q$ $\frac{R_p}{R_Q} = \frac{1}{0.25} = 4.0$
22	B	Potential difference across wire